



Indonesia Sovereign Asset Liability Management

Managing FX Exposure

Sovereign Debt Management Forum
December 3-4, 2014 , Washington DC

Background

● Rationale

- ▶ Asian Crisis 1997 Indonesia was very badly hit because of large exposure to short term FX Debt.
 - Small fx reserve not enough to service short term obligation
 - Large exposure to short term fx borrowing especially from private sector
 - Weak financial system without effective risk management could not prevent financial crisis
- ▶ Alarming signal about Increasing vulnerabilities to Public Balance Sheet due to higher growth in country FX borrowing especially by State Owned Enterprise.
 - Huge need for borrowing in FX to finance the expansion
 - Ample liquidity in global market provide easy access to borrow
 - Lack of natural hedge or hedged position
- ▶ 2008-2009 IMF-World Bank Mission to analyze the management of Financial assets and liabilities on the balance sheet of Government and Bank Indonesia, gave recommendations:
 - Coordinating the management of FX reserve and external debt portfolio in order to reduce currency risk
 - Strengthening institutional role between government and BI: reducing SBI (BI Certificates) role and optimizing the use of Government Securities as monetary instruments.
 - Improving cash management and coordination between debt & cash management units within MOF.

Cordination FX Management Current Condition

Bank Indonesia

Monetary Policy

Monetary Operation
to stabilize IDR exchange
rate

Managing FX Reserve
liquidity
safety
performance

BI has independency on
managing FX reserve

Government agent
bond auction
cash Payment
government Bank
received deposit

Operated Based on BI Law

Government

Fiscal policy

Managing fiscal (budget)

Collect tax and non tax
revenue, spending,
financing, investment

Issues goverment securities
both IDR and FX

Cordination with BI on
issuance plan

Managing debt portfolio

Securities and loan

Give onlending to SOE

State investment

Operated based on state
finance law and treasury law

State Own Enterprises

Agent development

Public service obligation

PT Pertamina, PT PLN, PT
Telkom

Financial service

Bank insurance

Pure Business

Goverment long term
investment

Received on lending from
government

Issue bond and loan both IDR
and FX,

Goverment permit for FX
borrowing

Operated Based on SOE Law

Surplus

IDR & FX
deposit

Subsidies,
govt. investment,
on lending

Divident, tax, royalty

2014 SALM new

Study on Managing FX Exposure

● Objectives

- ▶ Mapping the FX exposure conditions on:
 - Government., Bank Indonesia, and SOEs balance sheet. Consolidated B/S of Govt. and BI, and Consolidated of Neraca Pemerintah, BI, BUMN, dan Neraca Publik (Neraca Konsolidasi)
- ▶ Analyzing the vulnerabilities on FX exposures of Public B/S
- ▶ Results of Analysis will propose:
 - Portfolio management and its risk mitigation on public fx asset & liabilities
 - an alternative framework for FX management especially -for Government & BI B/S

● Scope

- ▶ Financial Assets & Liabilities of Government, Bank Indonesia and 6 SOEs
- ▶ Main currencies : Major and Significant Currencies (USD, JPY, and EUR)
- ▶ Risks:
 - At this time only mapping the net fx risk exposures on balance sheet of Government, BI, and SOEs
 - In future it will focus on analysis of interest rate and refinancing risks, NPV of assets & liabilities, analysis on duration/maturity of asset & liabilities, analysis on cash flow of public fx exposures

● Method

- ▶ Source of Data:
 - LKPP (Audited Govt. Financial Statement)
 - LKBI (Audited BI Financial Statement)
 - Annual Audited Financial Statement of 6 Non Banks – SOEs (incl. PT. Pertamina dan PT. PLN)
- ▶ Converting all the FX currencies into rupiah denominated numbers
- ▶ Applying assumptions for detailed breakdown of BI's balance sheet items

Balance Sheets of Govt., BI, SOEs

Government	2008	2013
<i>net exposures</i>	<i>1,502</i>	<i>2,247</i>
Assets	(135)	(124)
Demand Deposits in BI	(93)	(60)
On-Lending	-	(57)
Cash in BI	(42)	(7)
Liabilities	1,637	2,371
Bonds	906	1,661
Loans	730	710

Bank Indonesia	2008	2013
<i>net exposures</i>	<i>(397)</i>	<i>(774)</i>
Assets	(587)	(1,253)
Securities	(500)	(973)
FX Demand Deposits	(34)	(135)
SBN Ownership	(23)	(44)
Gold	(22)	(37)
Special Drawing Rights	(0)	(33)
FX Deposits	(7)	(31)
Foreign Currencies	(0)	(0)
Liabilities	190	480
Banks DD	85	323
Govt. DD	93	60
Term Deposits	-	57
SDR Allocations	4	37
FX Loans	7	3

SOEs	2008	2013
<i>net exposures</i>	<i>127</i>	<i>373</i>
Assets	(54)	(121)
Cash	(44)	(118)
Short term investment	(7)	(2)
Other Asset	(3)	(1)
Liabilities	181	493
Bonds	35	180
Loans	52	120
Trade Payables	46	85
Short Term Borrowing	11	61
On-Lending	30	40
Electricity Purchase Payable	8	7
Dividen to Govt.	15	0

Notes:

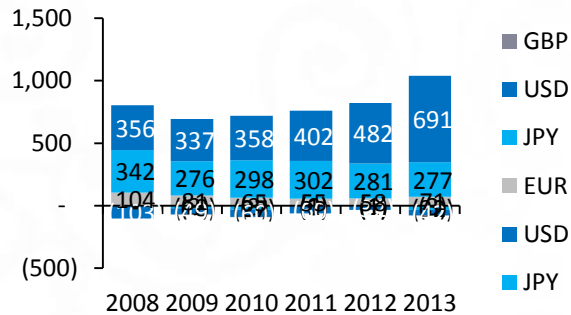
1. Nominal in trillion Rupiah,
2. Liabilities is shown by positive numbers and assets in negative ones,
3. Currencies includes all FX currencies and IDR
4. Govt.'s numbers are using LKPP 2008-2013 (audit)

- Government and SOEs had larger financial liabilities compared to their financial assets
- Bank Indonesia had larger assets than its liabilities
- Most of government financial asset is on Demand Deposits in BI and on the liabilities side most of it are government securities

- The largest asset in BI's balance sheet is the international reserves for the purpose of financing imports and debt services (from government and private sectors). The largest liabilities is the Demand Deposits for national banks
- The largest asset in SOEs balance sheet is in Cash and the largest liabilities are debts (bonds and loans)

Overview B/S - Government

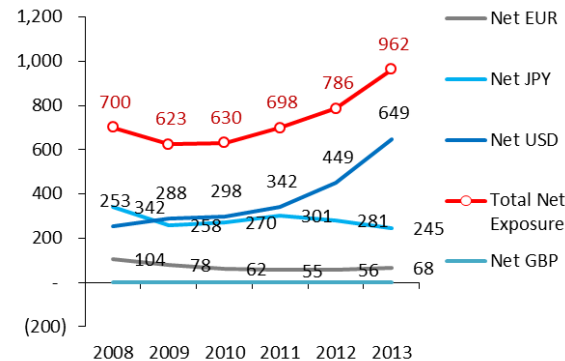
Govt.'s Gross FX Exposure



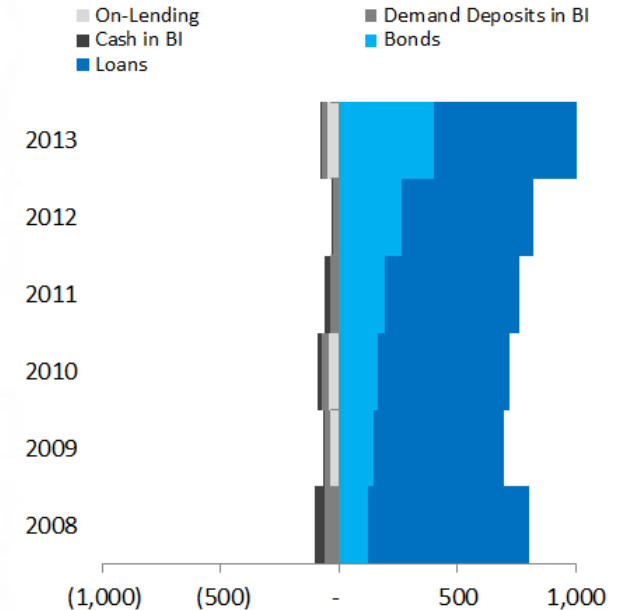
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Govt.'s Net FX Exposure



Govt.'s B/S Component



●Characteristic of B/S

►Assets :

- mostly in form of Demand Deposits in BI & On Lending to SOEs,
- mostly in fixed assets

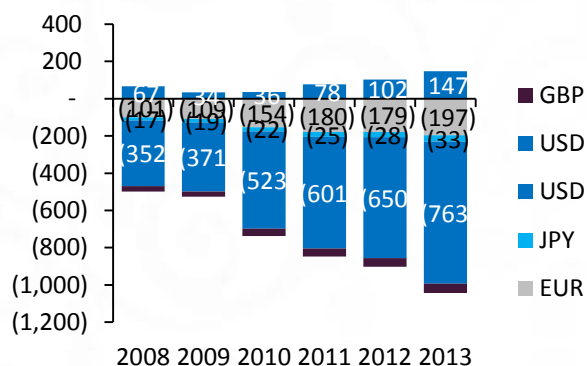
►Liabilities:

- Biggest portion is in FX Debt consists of FX Loans and Securities

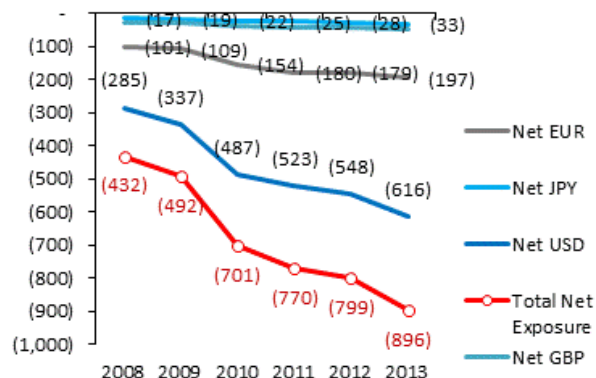
- In recent 5 years, the gross fx exposure in USD and EUR increased while JPY decreased. The net USD exposure in USD and JPY remained significant. However the net USD exposure increased sharply in 2013, because of widening budget deficit which caused USD borrowing to increase and Rupiah depreciation against USD

Overview B/S – Bank Indonesia

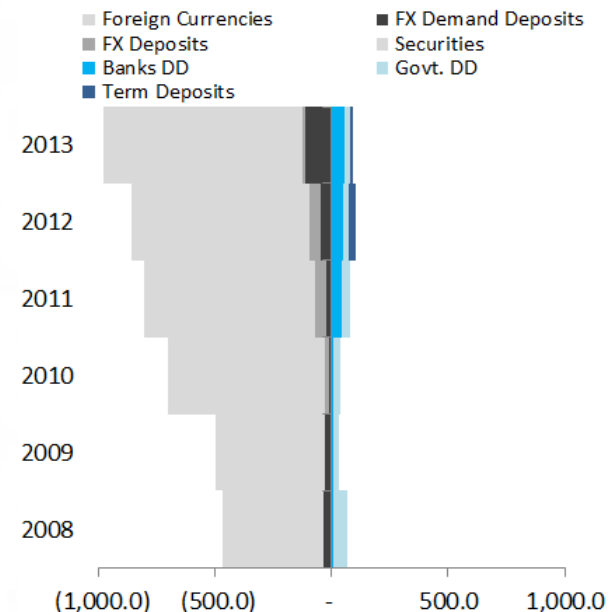
BI's Gross FX Exposure



BI's Net FX Exposure



BI's B/S Component



Notes:

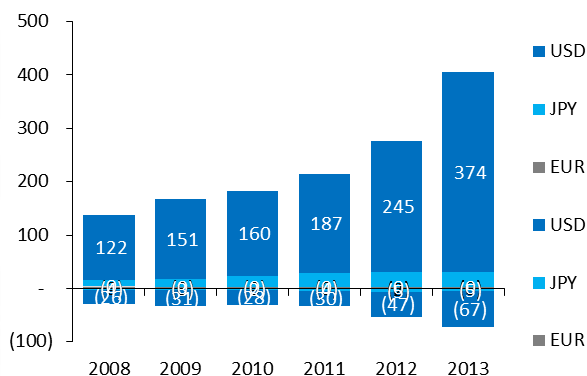
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- Characteristic of B/S
 - ▶ Assets: mostly in Securities (detailed currencies proportions are not clearly defined) and FX Demand Deposits
 - ▶ Liabilities: majorities in Bank Demand Deposits, Govt. Demand Deposits

- Fortunately on the BI side of B/S, their liabilities are much lower than their assets
- Those assets which are mostly in Securities (other sovereign bonds) increased dramatically after 2009, possibly because of the effect of large issuance Government's FX denominated bonds (as the consequences of increased budget financing).

Overview B/S - SOEs

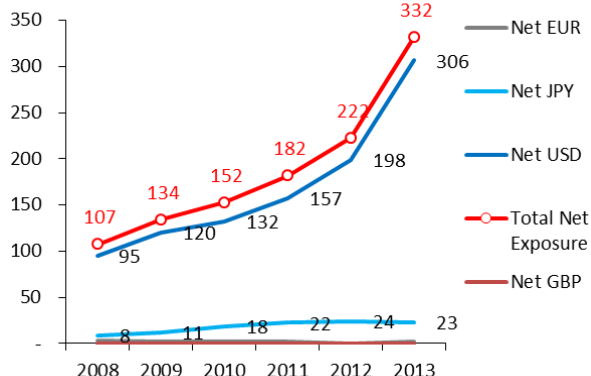
SOEs's Gross FX Exposure



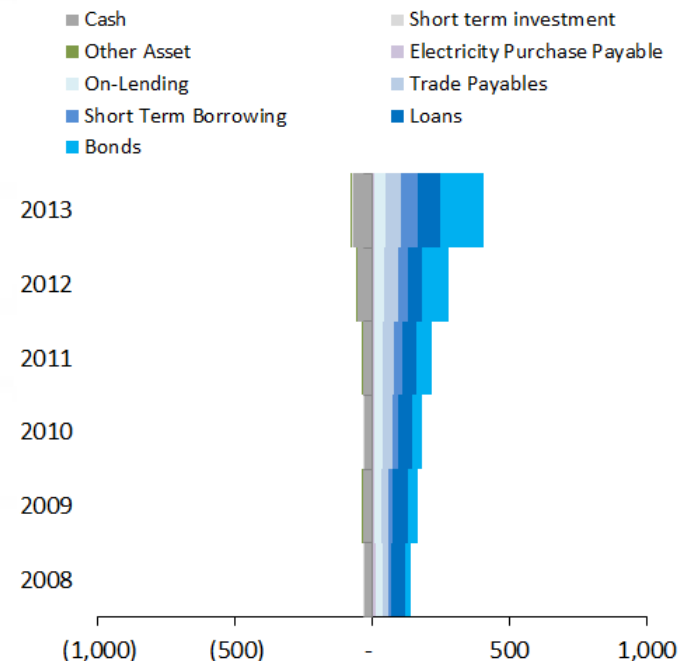
Notes:

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4. Numbers are using Financial Statements (audit) 2008-2013.

SOEs's Net FX Exposure



SOEs' B/S Components



● Scope

- Consists of 6 SOEs with largest assets and liabilities and also largest fx exposure
- 4 SOEs are public companies, the others are PT Pertamina and PT PLN
- These 6 SOEs can represent 80% of all SOEs (140 SOEs),
- Excluding SOEs' Banks, considering Regulations from BI for Banks to have less fx exposure in their portfolio, so we considered them to have very little net fx exposure

● Characteristic

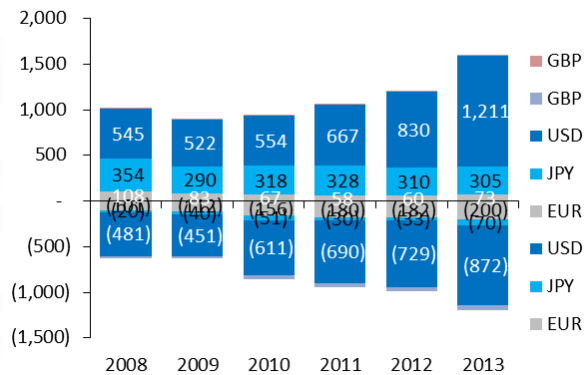
- Most of SOEs assets are in form of Inventory, Receivables from Govts, Cash
- Majorities of liabilities are in Loans, On-Lending, Trade Payables, and Bonds

● SOEs Liabilities

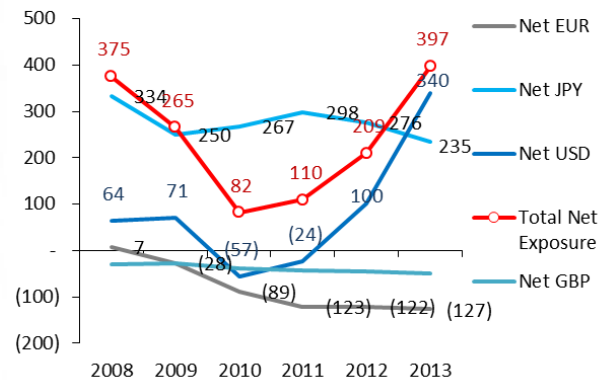
- The net positive fx exposure of USD had sharply increased after the it decreased in 2012
- Main contributors of the increasing net fx exposure of USD were from Pertamina's Debt (Rp122 trillion) and Short Term Borrowing (Rp61 trillion) and also PLN's Debt (Rp106 trillion)

Consolidated B/S

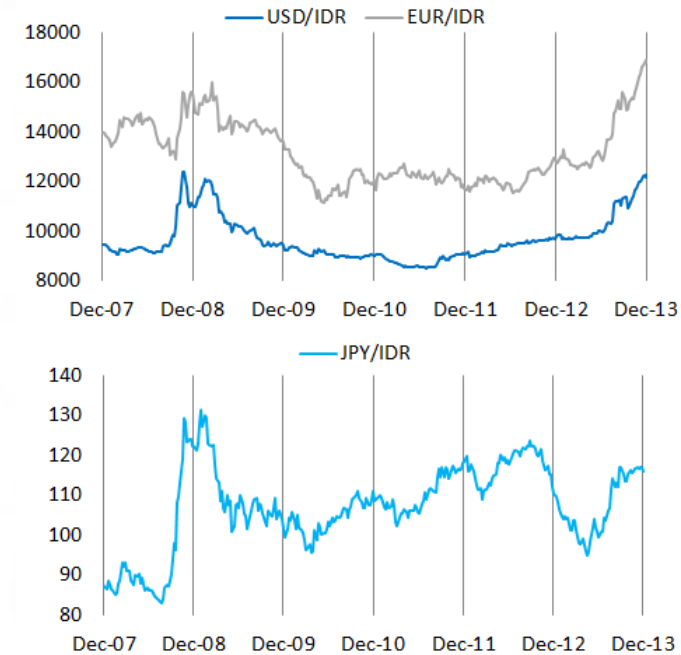
Govt. + BI + SOEs's Gross FX Exposure



Govt. + BI + SOEs's Net FX Exposure



FX to IDR Movement (2008-2013)



Findings

- Biggest Public FX Net Exposure is in USD then followed by JPY, EUR net exposure is negative (assets > liabilities)

Vulnerabilities Analysis

- Net exposures to USD is speeding up. Main reason is the widening fiscal deficit which translates into bigger financing need which causes increasing need for government FX borrowing. The other reason is the expansion of some SOEs especially PT. Pertamina to finance its foreign-investment and PT . PLN to meet its electricity capacity targets

- Net exposures JPY is slightly down, because even though the SOEs JPY debt is increasing, on the government side, JPY debt continues to decrease.

Conclusion & Recommendation

● Conclusion

▶ B/S of Government & BI

- Net FX Exposure JPY which is quite vulnerable → positive net exposure
- Net FX Exposure USD is within measurable/controllable level
- Net FX Exposure EUR provides more room to expand such as issuing EUR denominated Bonds . The EUR Bonds issuance will also support efforts to balance debt portfolio in JPY and USD denomination
- A need for coordination in high level officials to analyze the detail breakdown of FX reserves and to support strong and reliable analysis on Government & BI net FX exposures. (right now it is still challenging to gather detail data for each balance sheet due to regulation)

● Recommendations

▶ Increase fx asset

- Increase export repatriation → export revenue is not directly transferred to Indonesian economy

▶ Reducing fx debt issuance by reducing budget deficit

- Increasing issuance of domestic SBNs → need domestic market development to increase domestic market absorption
- Limiting issuance of fx denominated SBNs → SOMETHING MISSING HERE?
- Limiting fx borrowing for SOEs → need more coordination within government institutions

▶ Converting EUR assets to USD or JPY to minimize USD and JPY exposures

▶ Conduct hedging to mitigate USD & JPY risks

- Coordination with Bank Indonesia to map the exact amount of hedging. This need joint policy to determine which currency and how much exposure to be hedged, and who should do it.

Next steps on Managing FX Exposure

Next Step :

- Improving the research with more detail aspect among others about short term Horizon of FX exposure where currently not yet addressed and interest rate exposure aspect of consolidated balance sheet.
- Revising current regulation to give umbrella to a much better coordination as follow :
 - ▶ BI Law where new draft incorporate a clause about Sovereign ALM where the current is not.
 - ▶ Improve regulation about SOE FX borrowing which currently regulate a procedure for SOE to borrow in global market.
 - ▶ Create an MOU between Government and BI to arrange coordination mechanism on sovereign ALM including FX management.
- Create a new unit in ministry of Finance that in charge of managing sovereign balance sheet including FX exposure.
- Improve Coordination in fx debt and reserves composition
 - ▶ Realignment which will require more transparencies on detailed currency selection on some B/S item
 - ▶ Coordination in hedging policy to minimize amount required/executed for each hedging transaction.



Thank You



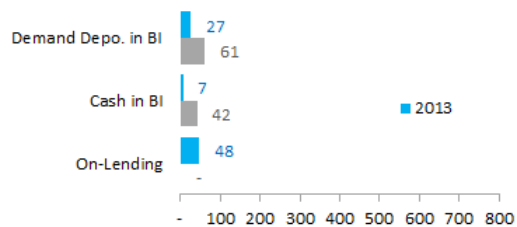
Additional Charts

B/S Compositions of Govt, BI, and SOEs in 2008 & 2013

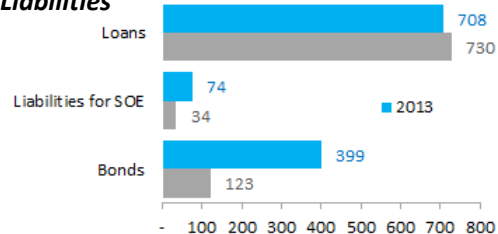
including all fx currencies

Government

Assets

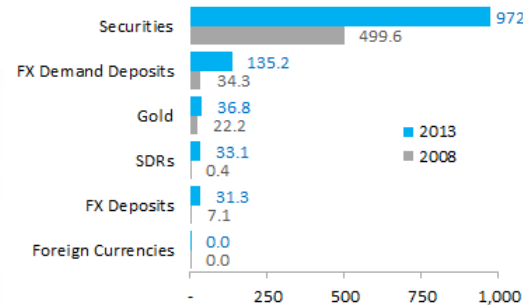


Liabilities

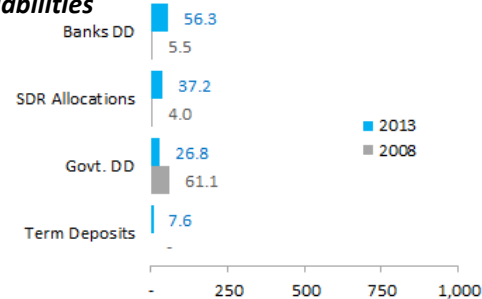


Bank Indonesia

Assets

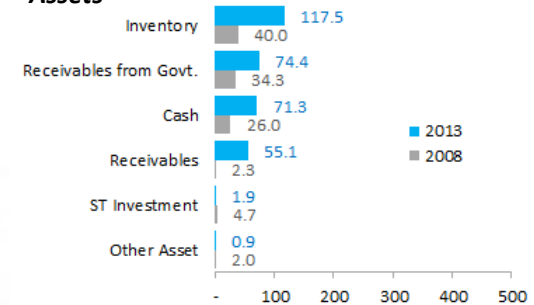


Liabilities

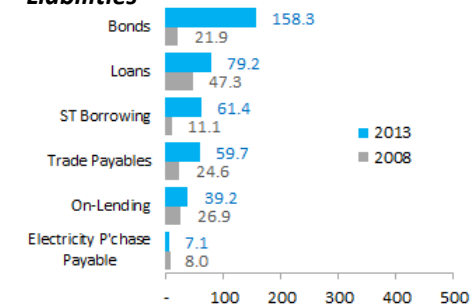


SOEs

Assets



Liabilities

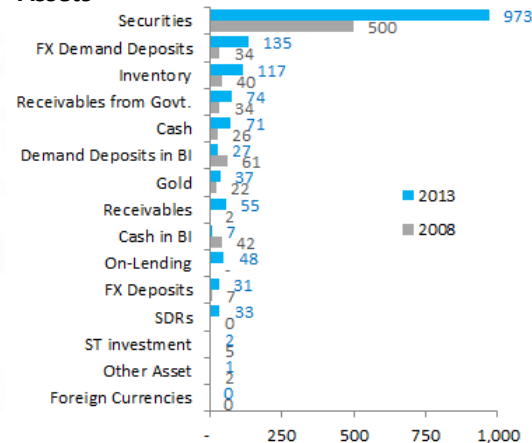


Compositions of Consolidated B/S in 2008 & 2013

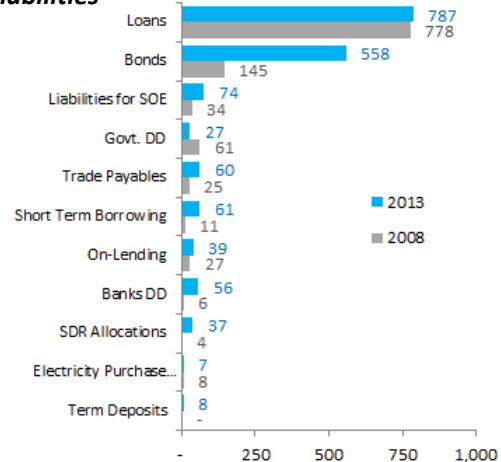
including all fx currencies

by B/S Components

Assets

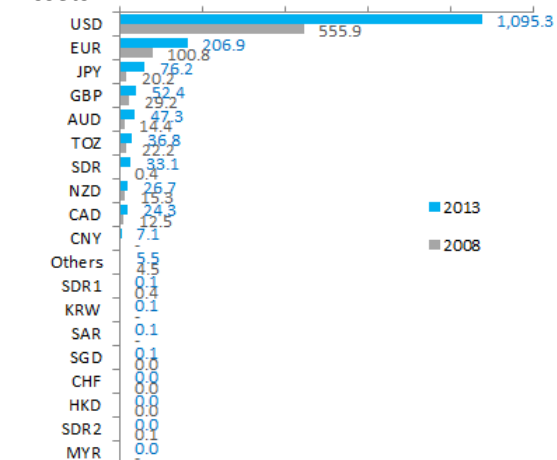


Liabilities

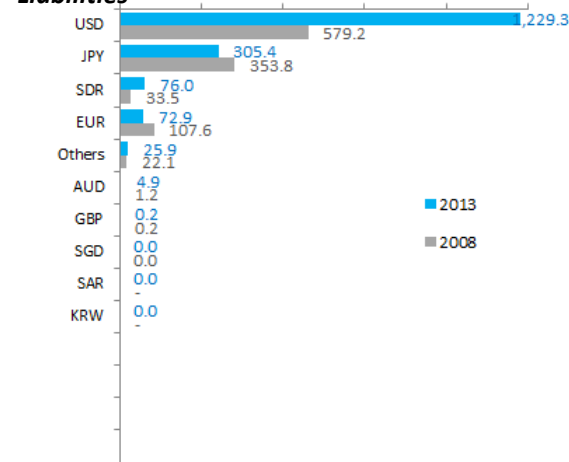


by Currencies

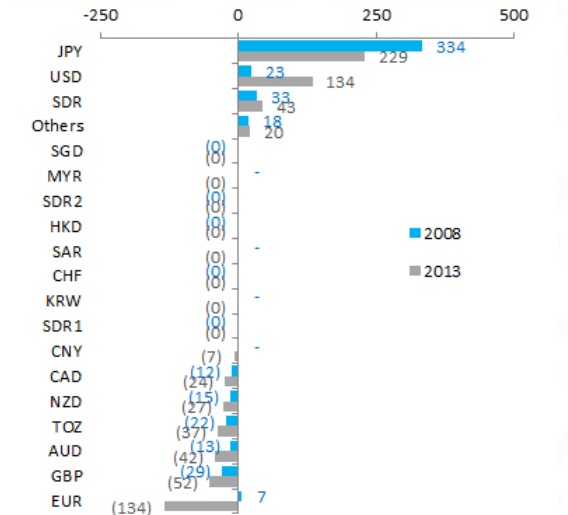
Assets



Liabilities



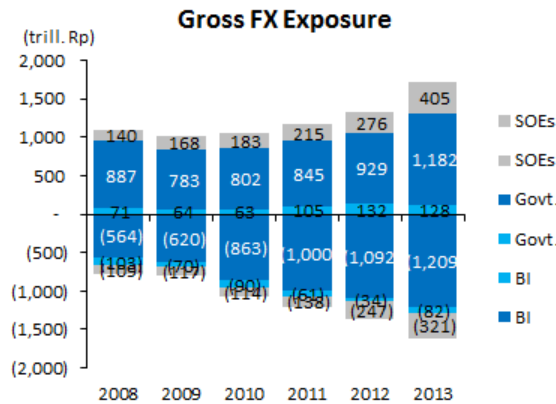
All Currencies Net Exposures



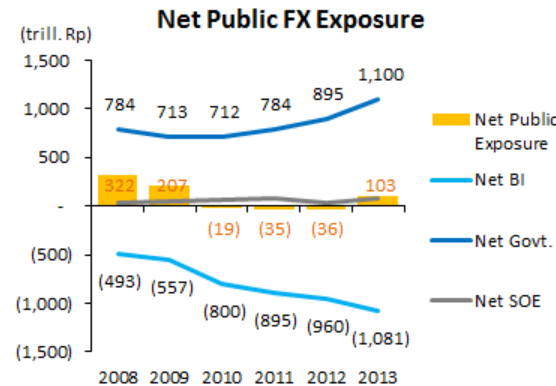
FX Exposures on Consolidated B/S

including all fx currencies

Gross Exposure



Net Exposure



Notes:

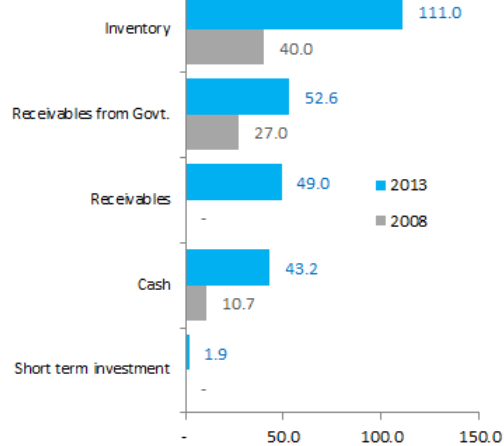
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- If considering all fx currencies, the public B/S experienced a deficit in net fx exposure
- This happened as a result of extensive fx borrowing conducted by Govt. (for financing budget deficit) and SOEs (for expansion) in 2013, while the size of BI's assets did not expand significantly

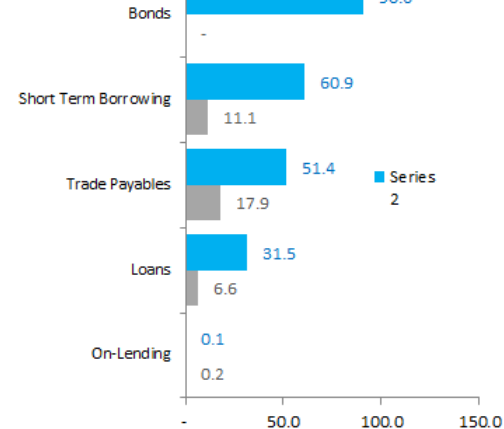
Sample of SOEs Balance Sheet

PT.Pertamina

Assets

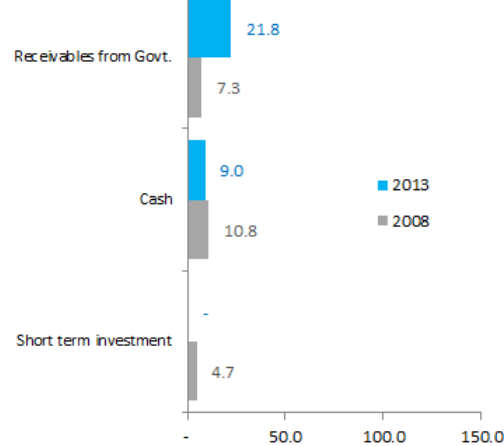


Liabilities

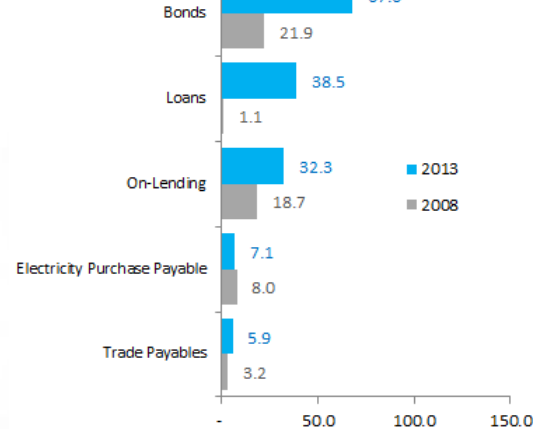


PT. PLN

Assets

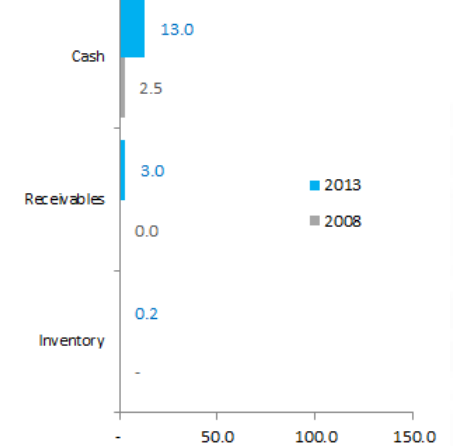


Liabilities



PT. PGN

Assets



Liabilities

